

Dinghao Luo

Neuroscientist | Applied Statistics, Experimental Design & Biomedical Data
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Open to relocation in Europe · Available January 2027; flexible after October 2026 PhD defence

Neuroscience PhD candidate with six years of experience designing controlled studies and analysing high-dimensional biological and behavioural data. Built reproducible Python pipelines for 50+ TB of electrophysiological, imaging, and behavioural data and applied statistical methods to test competing biological mechanisms.

CORE SKILLS

Programming	Python (pandas, NumPy, SciPy, statsmodels, scikit-learn), SQL (PostgreSQL), R, MATLAB
Statistics	GLMs, matching, survival analysis, repeated-measures designs, permutation and resampling inference, power analysis, multiple-testing correction
Data & software	Data validation, time-series analysis, Git, automated testing, documentation

EXPERIENCE

Graduate Researcher · Max Planck Florida Institute for Neuroscience May 2021–Dec 2026

- Designed controlled studies across 100+ animal recording sessions using electrophysiology, optogenetics, pharmacology, two-photon imaging, and behavioural tasks; defined outcome measures and analysed repeated measurements.
- Built reproducible Python pipelines processing 50+ TB of behavioural, electrophysiological, and imaging data, including data validation, event alignment, feature extraction, time-series analysis, and dimensionality reduction.
- Used GLMs, matching, and permutation inference to test competing biological mechanisms; rejected the original hypothesis when its predicted behavioural effect did not appear, then open-sourced and documented the analysis workflows for a first-author manuscript submitted to *Cell*.

Undergraduate Researcher · University of Cambridge, Paulsen Lab July 2019–Apr 2020

- Designed whole-cell electrophysiology experiments testing how dopamine changes synaptic-plasticity thresholds in hippocampal CA1; First-Class thesis.

Research Intern · UCL Queen Square Institute of Neurology July–Aug 2018

- Quantified cathepsin B activity in Down syndrome mouse models; co-authored a peer-reviewed paper in *Acta Neuropathologica Communications* (2023).

SELECTED PROJECTS

FibreSight · github.com/dinghaoluo/fibre-sight

- Built and open-sourced a PyTorch U-Net and Qt application for axon segmentation in microscopy images; achieved 0.93 recall and 0.83 F_2 on nine held-out sessions, reducing curation time from days to hours.

Kairos · github.com/dinghaoluo/kairos

- Built a Python/OpenAlex classification pipeline with reviewed labels, lagged features, and expanding-window validation; temporal baselines showed that calendar time explained most of the apparent predictive signal.

EDUCATION

PhD Candidate, Systems Neuroscience · FAU / Max Planck Florida Institute for Neuroscience 2021–2026

- IMPRS Fellow; defence scheduled for October 2026.

BA Biological Natural Sciences, First-Class Honours · University of Cambridge 2017–2020

- Jardine Scholar (full merit funding); Bundy Scholarship.

OTHER

Translator of *Entangled Life* (Sheldrake, 2020), Chinese edition 《菌络万象》; Pingshan Book Award 2025.
Science writer and former editor at *Scientific American China* (2020–2021) and *Neu-Reality* (2017–2021).
Native Cantonese and Mandarin; fluent English; conversational German.